CLI (Command Line Interface)

For the **CLI**, you can provide a Python script that renames a file via command-line arguments. Here's how you could implement it:

Python Code

```
import os
import sys

def rename_file(old_name, new_name):
try:
    os.rename(old_name, new_name)          print(f"File
renamed from {old_name} to {new_name}")      except
FileNotFoundError:    print(f"Error: {old_name} not found.")
except Exception as e:     print(f"An error occurred: {e}")

if __name__ == "__main__":
if len(sys.argv) != 3:
    print("Usage: python rename_file_cli.py <old_filename> <new_filename>")
else:
    rename_file(sys.argv[1], sys.argv[2])
```

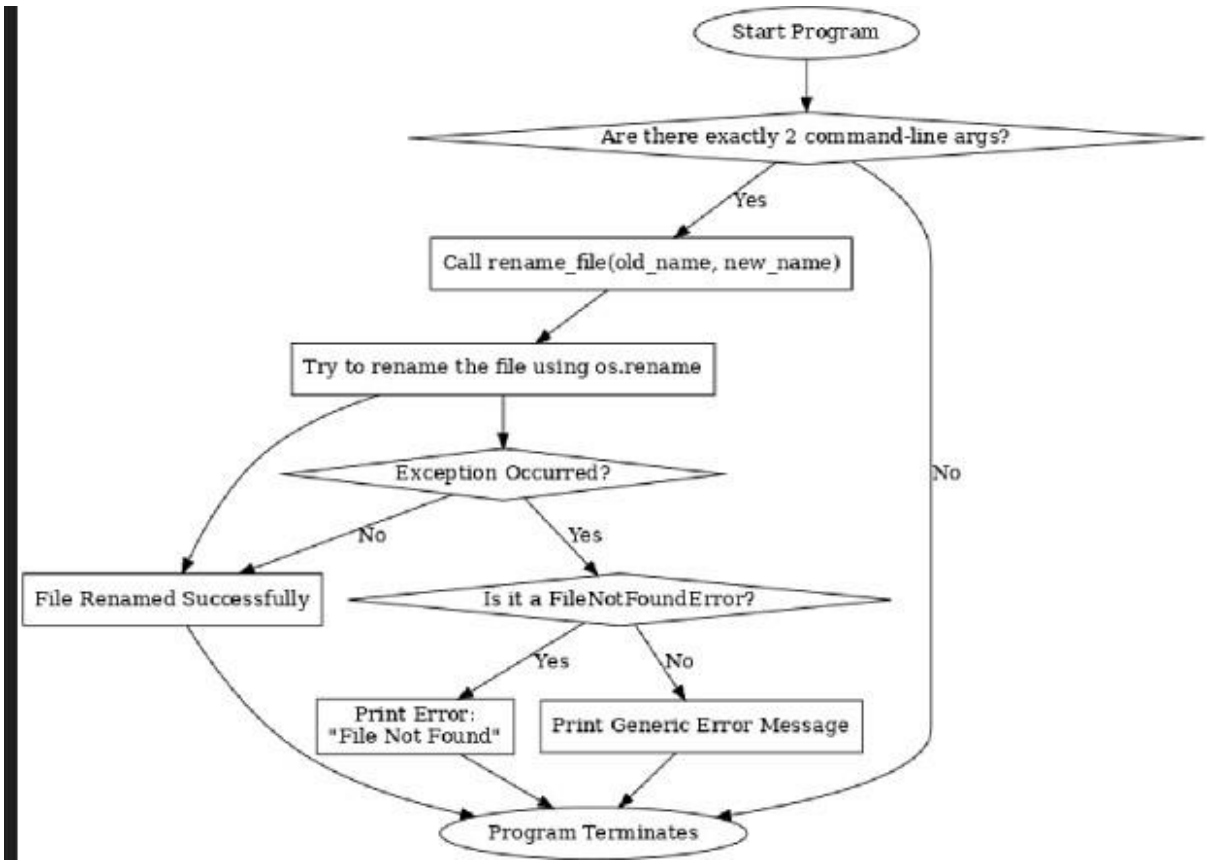
How it works:
The script accepts command-line arguments for the old and new filenames and renames the file.

How It Works:

1. **Imports:**
   - `os` for interacting with the file system.
   - `sys` for accessing command-line arguments.
2. **Function `rename_file`:**
   - Takes two arguments: `old_name` (current filename) and `new_name` (desired new filename).
   - Tries to rename the file using `os.rename()`.
   - Handles `FileNotFoundError` if the original file doesn't exist. ○ Catches any other unexpected exceptions.

To use it:

```
python rename_file_cli.py oldfilename.txt newfilename.txt
```



GUI (Graphical User Interface)

In the **GUI**, we will use **Tkinter** to create a simple interface where the user can type the old and new filenames.

```
import tkinter as tk from tkinter
import messagebox import os

def rename_file():
   old_name = old_filename_entry.get()
   new_name = new_filename_entry.get()

try:
      os.rename(old_name, new_name)        messagebox.showinfo("Success", f"File
renamed from {old_name} to {new_name}")     except FileNotFoundError:
      messagebox.showerror("Error", f"File {old_name} not found.")
except Exception as e:
      messagebox.showerror("Error", f"An error occurred: {e}")

# Set up the main window root
= tk.Tk()
```

```
root.title("File Renamer")

# Create and place labels, entries, and buttons tk.Label(root,
text="Old Filename:").grid(row=0, column=0) tk.Label(root,
text="New Filename:").grid(row=1, column=0)

old_filename_entry = tk.Entry(root)
old_filename_entry.grid(row=0, column=1)

new_filename_entry = tk.Entry(root)
new_filename_entry.grid(row=1, column=1)

rename_button = tk.Button(root, text="Rename File", command=rename_file)
rename_button.grid(row=2, columnspan=2)

# Start the Tkinter event loop root.mainloop()
```

How it works:
The GUI allows the user to input the old and new filenames, then click a "Rename File" button. Upon clicking the button, it attempts to rename the file and provides feedback via a message box.

Visual Layout:

- **Window Title:** *File Renamer*
- **Size:** 400x150 pixels (fixed, not resizable)

▯ Old Filename:

(Entry field aligned to the right with enough space to type file names)

▯ New Filename:

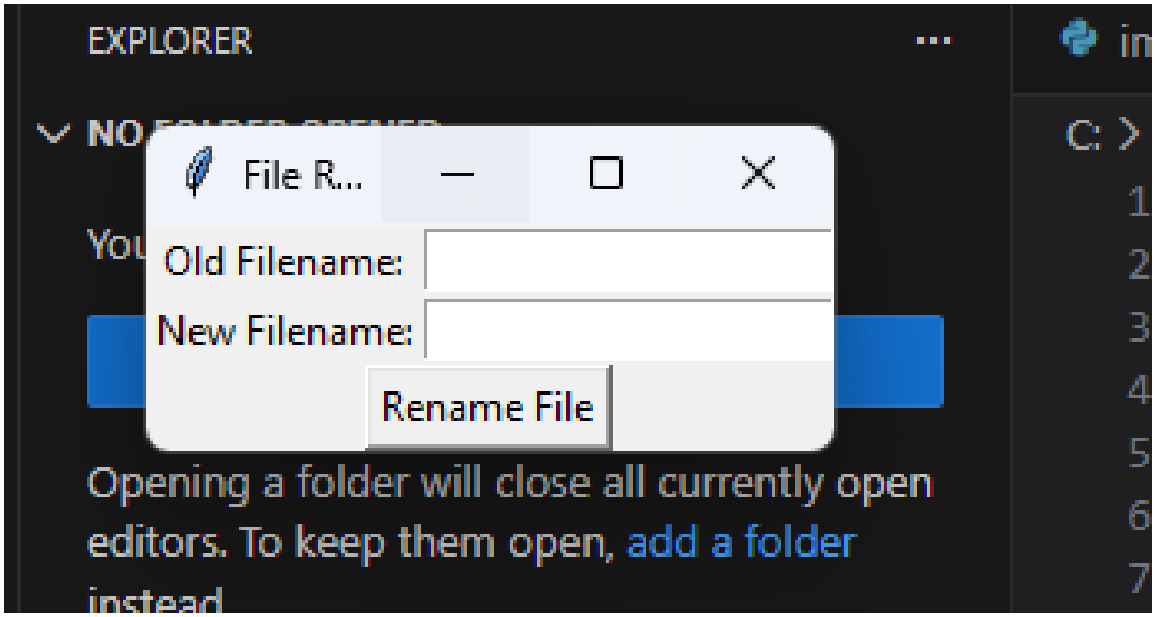
(Another entry field right below the first one) ✅

Rename File Button:

- **Color:** Green background with white text
- **Text:** *"Rename File"*
- **Centered below the input fields**

When you click the **Rename File** button:

- A pop-up will appear:
  - **Success Message:** Green tick icon with a confirmation.
  - **Error Message:** Red cross icon if there's an issue. ○ **Warning:** Yellow triangle if inputs are empty or overwriting a file.



VUI (Voice User Interface)

For the **VUI**, we can use the **SpeechRecognition** library to listen for voice commands. The user will say something like, "Rename oldfilename.txt to newfilename.txt."

```
import speech_recognition as sr import
os
```

```
def rename_file_from_voice_command(command):
   # Extracting old and new filename from the command
try:
      words = command.split(" ")
old_name = words[1]
      new_name = words[3]

      os.rename(old_name,  new_name)                print(f"File
renamed  from  {old_name}  to  {new_name}")          except
Exception as e:        print(f"Error: {e}")

def listen_for_command():
recognizer = sr.Recognizer()
   mic = sr.Microphone()

   print("Listening for command to rename a file...")
with mic as source:
recognizer.adjust_for_ambient_noise(source)
      audio = recognizer.listen(source)

try:
      command = recognizer.recognize_google(audio)
print(f"Command received: {command}")
rename_file_from_voice_command(command)     except
sr.UnknownValueError:
      print("Sorry, I couldn't understand the command.")
except sr.RequestError as e:
      print(f"Could not request results from Google Speech Recognition service; {e}")

if __name__ == "__main__":
listen_for_command()
```

How it works:

- The script listens for voice input using the microphone and recognizes the command using Google's speech recognition engine.
- The user can say something like: "Rename oldfilename.txt to newfilename.txt." The script extracts the filenames and attempts to rename the file.

Installation:

Ensure you have the necessary libraries installed:

```
pip install SpeechRecognition
pip install PyAudio  # For microphone input
```

User Satisfaction Comparison

To assess user satisfaction, you can compare the interfaces on the following aspects:

1. **Efficiency** (Time to complete task):
   - **CLI**: Most efficient for advanced users. ○ **GUI**: Moderate efficiency, dependent on visual navigation. ○ **VUI**: Dependent on speech recognition accuracy and environmental conditions.
2. **Learning Curve**:
   - **CLI**: Steep for non-technical users. ○ **GUI**: Easy to learn for most users.
   - **VUI**: Very shallow, as it mimics natural language commands, but voice accuracy issues might hinder this.
3. **Error Handling**:
   - **CLI**: Minimal feedback, error prone for typos. ○ **GUI**: Provides visual feedback with error messages. ○ **VUI**: Dependent on recognition; can misunderstand commands if not clear.

4. **User Satisfaction**:
   - **CLI**: Likely to be rated low by non-technical users, but advanced users will be very satisfied.
   - **GUI**: Likely to be rated highly by most users due to ease of use.
   - **VUI**: Satisfaction depends on voice recognition accuracy; users may find it convenient for hands-free tasks, but errors can cause frustration.

How to Measure User Satisfaction

You can implement a simple survey to measure user satisfaction after each task completion. For example:

```
def survey():    print("Rate your satisfaction with the following
interfaces (1-5):")
   cli_satisfaction = int(input("CLI (1-5): "))
gui_satisfaction = int(input("GUI (1-5): "))
   vui_satisfaction = int(input("VUI (1-5): "))

   print(f"\nYour satisfaction ratings:")
print(f"CLI: {cli_satisfaction}")    print(f"GUI:
{gui_satisfaction}")
   print(f"VUI: {vui_satisfaction}")
```

Final Thoughts

- **CLI** is best for power users who need speed and control.
- **GUI** is ideal for most users due to its intuitive nature and ease of use.
- **VUI** provides a hands-free experience, though it may be prone to inaccuracies